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Your LLM Evaluation Is *Lying* to You

How eval numbers trick us when measuring LLM quality, and the cheap habits that catch it.

Irem Ergun

Senior Machine Learning Researcher, Post-training · Cohere

ABOUT ME

Irem Ergun

Senior Machine Learning Researcher on the post-training team at Cohere. I work on the side of the API most talks never see.

Ships models

Contributed to Command R+, Command A, and Command A+.

Publishes papers

NeurIPS, ACL, IEEE

Global talent in AI research

Endorsed by the Royal Society of the UK



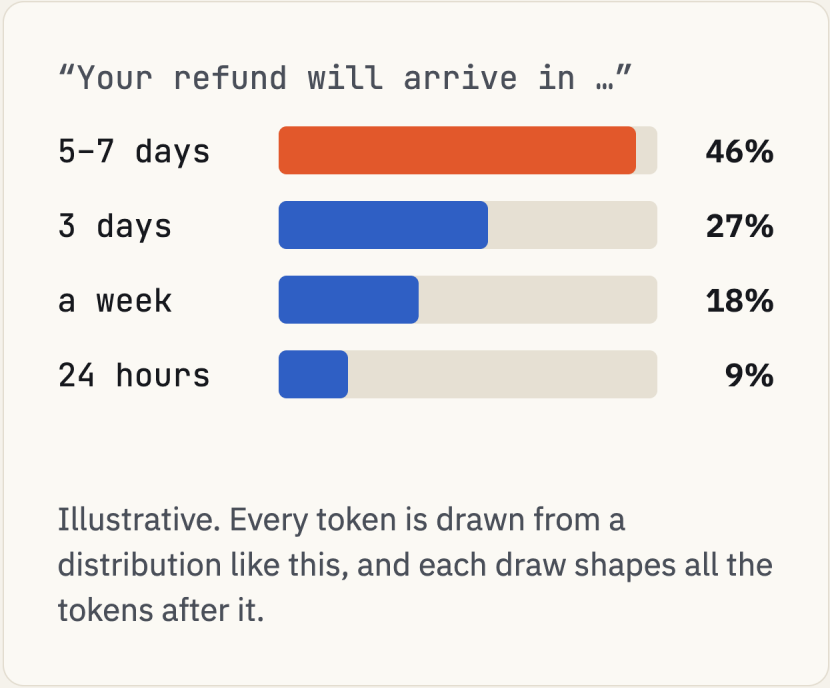
THE GAP

Everyone here can ship an LLM feature. Far fewer can prove it works.

This talk: four lies your eval tells you, and how to stop believing them.

WHY EVAL?

Your model is a distribution, not a function



Illustrative. Every token is drawn from a distribution like this, and each draw shapes all the tokens after it.

80 / 1,000 different answers from the same prompt at temperature 0

76 pts accuracy swing from meaning-preserving format changes

84% → 51% GPT-4 on the same questions, three months apart

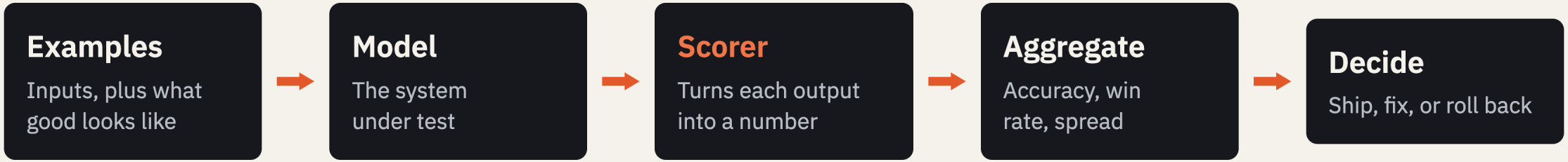
You can’t test a distribution by trying it once. You have to measure it.

Your LLM Evaluation Is Lying to You

He & Thinking Machines 2025; Sclar et al. 2024; Chen et al. 2024

BEFORE THE LIES

Every eval is the same loop



THE SCORER IS WHERE EVALS DIFFER, FROM RIGID TO FLEXIBLE

Exact match and regex

Labels, numbers, JSON fields. Free and repeatable.

Reference overlap

BLEU, ROUGE: word overlap with a human answer. Misses meaning.

LLM-as-a-judge

A model grades against a rubric. Flexible, cheap, biased.

Human review

The reference for judgment. Slow and expensive.

A SHORT HISTORY

How we got here

2002

Reference overlap

BLEU counts n-gram overlap with a human reference. ROUGE follows in 2004.

2018

Benchmark suites

GLUE rolls many tasks into one leaderboard number.

2020–21

Checkable answers

MMLU multiple choice, GSM8K answers pulled out by regex, HumanEval code run against unit tests.

2023

LLM-as-a-judge

MT-Bench and Chatbot Arena: strong judges agree with human preferences over 80% of the time.

2024+

Agents

Score the whole trajectory: tools, steps and outcome.

2026

Typed decisions

Models like Jev answer with probabilities instead of text.

Each step fixed the last one’s blind spot, and opened up a new way for confusion.

ROADMAP

Four lies, four fixes

01	*“It beats the big model on the benchmark.”*	THE BENCHMARK LIES
02	*“The LLM judge said it improved.”*	THE GRADER LIES
03	*“My agent works. It called all the tools.”*	THE TRAJECTORY LIES
04	*“It works.”*	YOU LIE TO YOURSELF

LIE 01 · THE BENCHMARK

“It beats the big model on the benchmark.”

WHY IT’S FALSE

Saturation. Once top models cluster above ~90%, the gaps between them are noise. Nearly half of 60 benchmarks studied in 2026 are saturated.

Contamination. Test data leaks into training. The Llama 2 report found over 16% of MMLU samples contaminated.

THE FIX

Ask how much of the score survives decontamination. Prefer live benchmarks such as LiveBench, and independent runs such as Artificial Analysis. Best of all: 20 examples from your own failures.

Touvron et al. 2023; Akhtar et al. 2026; White et al. 2025

LIE 02 · THE GRADER

“The LLM judge said it improved.”

WHY IT’S FALSE

It won on cost. Evaluating 1,000 outputs: about $500 with crowdworkers, about $3.60 with an LLM.

It has biases. Position, verbosity and self-preference. GPT-4 favored its own answers with a ~10% higher win rate.

It depends on the judge. Across 13 judge models, only the largest aligned reasonably with humans. Judges in alignment can still assign vastly different scores.

THE FIX

Calibrate against human labels before you trust it. Randomize answer order. Spot-check a sample every release.

Zheng et al. 2023; Kasner et al. 2026, Thakur et al. 2025

LIE 03 · THE TRAJECTORY

“My agent works. It called all the tools.”

WHY IT’S FALSE

Right calls, wrong outcome. An agent can call every tool correctly and still fail the task.

Right outcome, wrong path. Or it succeeds with the wrong tools, bad inputs, needless loops and a burned token budget.

THE FIX

Score the whole trajectory, not just the final answer.

Survey on Evaluation of LLM-based Agents, arXiv:2503.16416

LIE 03 · THE FIX

Score the trajectory, not just the answer

Tool correctness
Right tool, right arguments, at every step.

Task completion
Did the user’s goal actually get done?

Step efficiency
No loops, no detours, sane token spend.

Plan adherence
Did it follow the plan it made?

A THREE-LEVEL PIPELINE

1 Unit tests on tool calls → **2** LLM judge on trajectories → **3** Online eval on 5 to 10% of traces

Yehudai et al. 2025; MCP-Atlas (Bandi et al. 2026); Toolathlon (Li et al. 2025)

LIE 04 · YOU

"It works." Three quiet self-deceptions

The lie	Why it fools you	The fix
"It worked on my examples."	Tiny n, and the same examples you tuned on	A held-out set from real usage
"The number went up."	One run of a non-deterministic model	Several runs, report the spread
"It's great in English."	Quality drops outside English, worst for non-Latin scripts	Evaluate in the languages you serve

Ahuja et al. 2023 (MEGA); 2024 (MEGAVERSE)

```python
def score_one(pred, gold):
    if pred is None:
        return 0, "invalid_json"
    if pred.get("category") not in CATEGORIES:
        return 0, "off_schema"
    if pred.get("priority") not in PRIORITIES:
        return 0, "off_schema"
    ok = (pred["category"] == gold["category"]
          and pred["priority"] == gold["priority"])
    return (1, "correct") if ok else (0, "wrong")

def run_once(data):
    correct = 0
    for row in data:
        pred = parse(call_model(row["message"]))
        s, why = score_one(pred, row["label"])
        correct += s
    return correct / len(data)

acc = run_once(load_dataset("dataset.jsonl"))
raise SystemExit(0 if acc ≥ 0.80 else 1)
```

BUILD IT

A real eval fits on one slide

1 Examples come from real traffic, labeled by hand.

2 A deterministic scorer that says why it failed.

3 A threshold and a non-zero exit, so it runs in CI.

Full code: https://github.com/irombie/llm-eval-lies

KNOW WHEN A NUMBER IS NOISE

Same eval, same model, five runs

20 pts

between the best and worst run

A change smaller than this spread is variance, not a result.



Mean 81.3%, stdev 7.8%

Mock model · python eval.py --mock --runs 5 --seed 1772

READ A BENCHMARK CRITICALLY

Five questions before you trust a number

1 **Is it saturated?**
Top models clustered above ~90% means the gaps are noise.

2 **How much survives decontamination?**
Contamination is a spectrum, and it varies by model.

3 **Who ran it, and on what harness?**
Agentic scores depend on the scaffold. Self-reported numbers need a second look.

4 **Is there a confidence interval?**
If neighbors' intervals overlap, the ranking is not settled.

5 **Does the task look like mine?**
A public benchmark score is not your-task quality.

ONE MORE THING · JEV BY TYPESAFE

Frontier intelligence, as a *function call.*

```
decide(state, typed questions) → answers + probabilities + confidence
```

70–500 ms
end-to-end per call

$0.042
per million input tokens, output free

0
type errors: every possible answer is defined in advance

Named after William Stanley Jevons: when intelligence gets cheaper, we find more uses for it.

TypeSafe launch post, Sep 15, 2026 (vendor figures)

ONE MORE THING · JEV BY TYPESAFE

Smart if-statements for agents

Route

Pick the model, tool or skill for each request. Weak matches get declined, not guessed.

Gate

Approve a bash command or file edit before it runs. Fail closed when unsure.

Browse

Choose the next click on a page. Hand the typing to an LLM.

Compact

Keep only the tool results that matter before they fill the agent's context.

Escalate

Triage tickets into queues, and send low-confidence cases to a human.

Play Doom

At 10 decisions a second, a demo bot plays Doom for about $7 an hour.

ONE MORE THING · JEV BY TYPESAFE

The LLM writes. Jev decides. Code acts.

LLM
Plans, reasons, writes text

→

Jev
Answers typed questions

→

Your code
Branches on the answer

`Choice`

"Which team should handle this ticket?"

billing · technical · account · unknown

`Score`

"How urgent is this ticket?"

Low → Medium → High → Critical. High and up pages on-call.

`Noul`

"This tool result helps answer the user's question."

Yes: keep it in context. No: drop it before the agent sees it.

Keep the control flow in your code, always offer an "unknown" option, and set confidence thresholds from your own data and error costs.

Your LLM Evaluation Is Lying to You

TypeSafe docs; AIsa guide; LiteLLM Jev guardrail

MONDAY MORNING

Five things to do this week

01 Build 20 eval examples from real failures.

02 Calibrate your LLM judge once against your own labels.

03 Score the agent’s trajectory, not just its answer.

04 Rerun before you claim a win, and report the spread.

05 Test in the languages your users actually speak.

Further reading

WHY EVAL

He & Thinking Machines Lab (2025). Defeating Nondeterminism in LLM Inference. thinkingmachines.ai/blog

Sclar et al. (2024). Quantifying Language Models' Sensitivity to Spurious Features in Prompt Design. ICLR. arXiv:2310.11324

Chen, Zaharia & Zou (2024). How Is ChatGPT's Behavior Changing over Time? Harvard Data Science Review.

MULTILINGUAL

Ahuja et al. (2023). MEGA: Multilingual Evaluation of Generative AI. EMNLP. arXiv:2303.12528

Ahuja et al. (2024). MEGAVERSE. NAACL. arXiv:2311.07463

BENCHMARKS AND CONTAMINATION

Akhtar et al. (2026). When AI Benchmarks Plateau. ICML. arXiv:2602.16763

White et al. (2025). LiveBench. ICLR. arXiv:2406.19314

Touvron et al. (2023). Llama 2. arXiv:2307.09288

Li et al. (2024). An Open-Source Data Contamination Report for LLMs. EMNLP Findings. arXiv:2310.17589

Xu et al. (2024). Benchmark Data Contamination of LLMs: A Survey. arXiv:2406.04244

Singh et al. (2024). Evaluation Data Contamination in LLMs. arXiv:2411.03923

Links to every paper: github.com/irombie/llm-eval-lies

Further reading

JUDGES

Zheng et al. (2023). Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. NeurIPS. arXiv:2306.05685

Thakur et al. (2025). Judging the Judges. GEM Workshop, ACL. arXiv:2406.12624

Kasner et al. (2026). LLMs as Span Annotators. arXiv:2504.08697

AGENTS

Yehudai et al. (2025). Survey on Evaluation of LLM-based Agents. arXiv:2503.16416

Bandi et al. (2026). MCP-Atlas. arXiv:2602.00933

Li et al. (2025). The Tool Decathlon (Toolathlon). ICLR 2026. arXiv:2510.25726

JEV

TypeSafe (2026). Introducing System One Models and Jev. typesafe.ai/blog

Li et al. (2026). JEV-as-a-Judge: Accept When Confident, Escalate When Unsure. arXiv:2609.26550

DeepEval docs. JevEval metric. deepeval.com/docs/metrics-jev-eval

HISTORY TIMELINE

Papineni et al. 2002 (BLEU); Lin 2004 (ROUGE); Wang et al. 2018 (GLUE); Hendrycks et al. 2020 (MMLU); Cobbe et al. 2021 (GSM8K); Chen et al. 2021 (HumanEval)

QUESTIONS

Teşekkürler. *Questions?*

Now go find out what your eval has been hiding.😶‍🌫️

Slides and code
github.com/irombie/llm-eval-lies

X
x.com/irombie

LinkedIn
linkedin.com/in/irem-machine-learning-engineer